

SecurePouch — Technical Product Specification

Smart Anti-Theft Travel Pouch

Status	Working tracker demo (June 2026) — Nano 33 BLE prototype paired to the Android app, relaying locations to the live dashboard end-to-end. Gen-1 FDM demonstrator enclosure ready.
Last updated	June 2026
Document purpose	Consolidated reference specification for hardware, software, and physical design decisions made to date

1 Product Overview

A smart anti-theft travel security pouch combining slash-resistant textile construction with integrated LTE-M/GNSS tracking, BLE proximity detection, biometric (fingerprint) access control, and an active audio-visual alarm system. Target users: safety-conscious travellers, international tourists, urban commuters, and concerned parents.

Business model: one-time device purchase plus cellular data subscription (Hologram eUICC).

Development roadmap: demo hardening → anti-theft features → nRF9151 DK prototype → peripheral drivers → PCB (see §15).

1.1 Confirmed Scope

- **Lock:** electromechanical — spring-loaded solenoid pin latch (fail-locked), with recessed mechanical key override.
- **Connectivity:** LTE-M (NB-IoT fallback), GNSS, and BLE 5.3. No Wi-Fi radio on-device.
- **Alarm:** piezoelectric diaphragm driven at resonance into a Helmholtz resonance chamber, plus an array of high-brightness (~ 1 W) strobe LEDs.
- **Enclosure:** rigid housing for electronics and latch, with a passport-sized fabric body. Final form factor is not decided; material expected to be machined aluminium. Gen-1 demonstrator is FDM-printed and ready.

1.2 Descoped Items

- RFID-blocking construction
- Waterproofing (IPX4 target removed)
- RF regulatory compliance work (FCC/CE) — deferred until commercialisation
- nRF54 / BLE Channel Sounding — rejected; useless without phone-side support. RSSI zones are the ranging ceiling; UWB would be the precision-finding route if ever needed
- Embedded push service in the Android app — proximity alerts are local notifications; pouch commands piggyback on the scan service's 30 s cycle (UnifiedPush/Sunup remains for FMD phone commands only)

2 Design Requirements

Acknowledged trade-offs: security vs. portability; subscription cost vs. cellular capability; RF regulatory compliance required before any commercial release.

Requirement	Notes
Portable & lightweight	Target total weight <150 g including electronics
Slash / stab resistant	Primary physical security requirement (para-aramid layer)
Easy to use	Single-tap arm/disarm; biometric or remote unlock
Global connectivity	LTE-M/NB-IoT on Hologram's 550+ carrier network, 190+ countries
End-to-end encryption	All device-server and server-client communication
Accessible without phone	Web dashboard functional from any browser
Fail-safe locking	Mechanical key override if electronics fail; fail-locked on power loss

3 Physical Design

3.1 Material Stack

Three layers of protection, outside to inside:

[500D Cordura PU nylon]	<- abrasion resistance, weather, aesthetics
[Aramid PARATEX 400P]	<- slash and stab resistance (structural core)
[Robic Diamond Ripstop]	<- interior finish, lightweight backing

Item	Part no.	Description	Supplier
Outer shell	40201.ANTHR	500D Cordura, PU-coated nylon, 240 g/m ² , anthracite, 1.5 m width	Extremtextil
Cut/stab layer	21-131	Aramid PARATEX 400P woven para-aramid, 150 cm width	TEXfire
Inner lining	73078.SW	Diamond Ripstop nylon, Robic Ocean Recycling, 70 den, UTS-coated, 100 g/m ² , 1.49 m width	Extremtextil
Sewing thread	71980.ROHGLB	Kevlar/aramid thread, size 75, 100 m	Extremtextil

Layer rationale:

- *500D Cordura (outer)*: more flexible than 1000D nylon; protects against scuffs, rain, and dirt; good sewability; shields the aramid layer from exposure.
- *PARATEX 400P (middle)*: woven para-aramid (comparable to 400 gsm Kevlar). Woven construction is more flexible and more cut-resistant than aramid felt; resistant to knife attack; acts as the structural core. Replaces the earlier UHMWPE (Dyneema) selection — equivalent or better stab resistance, good slash resistance, readily available cut-to-size.
- *Robic 70 den ripstop (inner)*: protects contents, gives a smooth interior, and reduces friction between the aramid layer and stowed items.

Tooling: Kretzer Finny TecX1 micro-serrated scissors, 8 cm (Extremtextil 70954) — required for clean cuts through aramid fabric.

3.2 Dimensions (Compact Variant)

- Length: ~155 mm (passport standard + margin)
- Width: ~98 mm
- Depth: TBD — pending final lock/enclosure assembly design
- Wearing options: neck/shoulder lanyard, cross-body strap, or jacket pocket

3.3 Enclosure

The electronics, latch, acoustic cavity, and fingerprint sensor are housed in a rigid enclosure integrated with the fabric body. The final form factor is **not decided**.

- **Gen-1 demonstrator:** FDM-printed — complete and ready.

- **Final material:** expected to be machined aluminium (CNC for the 5-unit prototype run), pending form-factor decision.
- **Constraints on the final design:**
 - Rigid faces at the latch interface to guarantee solenoid pin alignment.
 - BLE PCB trace antenna must face outward with no metal obstruction — a full-aluminium enclosure needs an antenna window or external antenna.
 - Helmholtz cavity with external aperture for the piezo diaphragm (§7.1).
 - Fingerprint sensor flush-mounted on an outer face.
 - Recessed key slot for the mechanical override.
 - PCB module serviceable without fabric disassembly.
- **Open concern — discretion:** whether the rigid enclosure reads as discreet travel gear or conspicuous hardware; evaluate against the FDM mock-up before locking dimensions.

3.4 Closure & Lock — Solenoid Pin Latch

The original zipper-based closure has been abandoned in favour of a rigid latch closed by a spring-loaded solenoid pin.

Latch mechanism (confirmed):

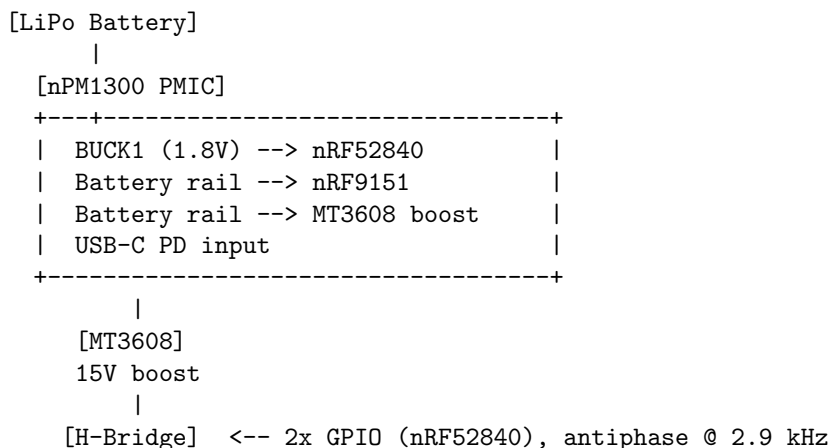
- Spring-loaded steel pin extends into a machined receiver hole in the mating latch face.
- Spring holds the pin extended: **fail-locked by default**. The solenoid retracts the pin briefly (~ 0.5 s) on an unlock event only.
- Drive: single GPIO + 2N7002 MOSFET; brief current pulse, no holding current in either state.
- Solenoid supply: 3.7 V battery rail or a small 5 V boost — to be selected empirically by retraction-force testing.
- Rigid latch faces guarantee pin alignment, eliminating the principal weakness of solenoid pins in flexible assemblies.

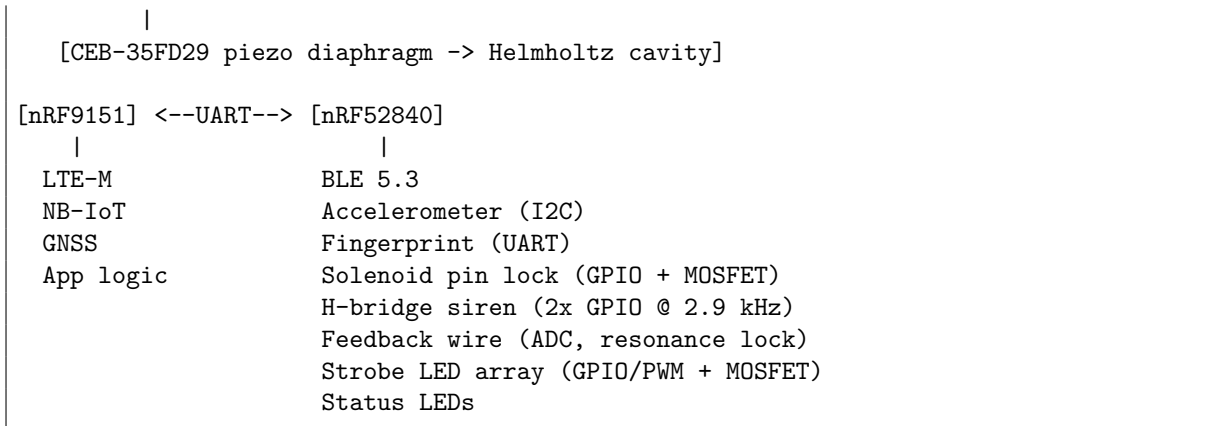
Rationale over servo cam (abandoned): the camming action of a servo lock exists to tolerate misalignment, which rigid latch faces make unnecessary. The solenoid pin is simpler (one moving part), smaller, lighter, inherently fail-locked, and needs only a GPIO pulse rather than PWM.

Mechanical fallback — recessed key slot: a keyhole manually drives a push rod that retracts the pin. Operates with zero power. The slot is recessed into the enclosure body and inaccessible without the correct key.

4 Electronic Architecture

4.1 System Block Diagram





4.2 Core ICs

Component	Part	Function
App MCU + LTE-M/GNSS	Nordic nRF9151-SICA	Cortex-M33 @ 64 MHz; LTE-M/NB-IoT modem; integrated GNSS; PSM sleep $\sim 2.5 \mu\text{A}$
BLE SoC + peripheral MCU	Nordic nRF52840	Cortex-M4F @ 64 MHz; BLE 5.3; 256 KB RAM; 1 MB flash; drives all local peripherals. Currently prototyped on an Arduino Nano 33 BLE
PMIC	Nordic nPM1300-CAAA-R	USB-C PD charging (up to 1.5 A); fuel gauge; BUCK1 (1.8 V $\rightarrow$ nRF52840); battery rail (3.7 V $\rightarrow$ nRF9151)
Accelerometer	Bosch BMA400	Triaxial; I ² C; $<1 \mu\text{A}$ wake mode; hardware motion-wakeup interrupt; 2 $\times$ 2 mm
Boost converter (siren)	MT3608 (C84446)	Battery $\rightarrow$ 15 V rail for H-bridge (30 V p-p across piezo)

Inter-MCU communication: bidirectional UART. The nRF9151 is the primary application processor and LTE-M/GNSS hub; the nRF52840 handles BLE proximity, local sensor polling, and all actuator control. The UART message protocol (status, commands, GNSS handoff) is not yet defined (roadmap P3).

5 Connectivity

5.1 Cellular

- Technology: LTE-M primary, NB-IoT fallback; modem integrated in nRF9151.
- SIM: Hologram Hyper eUICC, nano (4FF); 190+ countries, 550+ carrier networks; OTA profile management.
- Power saving: PSM + eDRX; target minutes-long modem sleep between location pings.

5.2 GNSS

- Receiver integrated in nRF9151; constellations: GPS, GLONASS, Galileo, BeiDou.
- A-GNSS via nRF Cloud for <5 s cold start.
- Fallback: cell-ID and Wi-Fi positioning via nRF Cloud Location Services (network-side; no Wi-Fi radio on-device).

5.3 BLE

- Radio: nRF52840, BLE 5.3 (BLE-only; no Classic).
- Primary uses: tracker relay advertising (§12.2) and dead-man switch — alarm triggers when the link to the user’s phone is lost beyond a configurable proximity threshold.
- Antenna: PCB trace inverted-F; no discrete component; requires copper keep-out zone.

Ranging realism: single-phone RSSI gives distance zones (near/medium/far/gone), not coordinates. The dashboard draws the pouch as a radius ring around the phone; the dead-man alarm uses an RSSI threshold plus disconnect detection with a ~ 30 s grace delay.

5.4 Antennas

Antenna	Part	Notes
Cellular + GNSS (2-in-1)	Taoglas FXP301	600 MHz–8 GHz cellular + GNSS L1; 105×20×0.24 mm flex PCB; peel-and-stick; dual IPEX/U.FL
BLE	PCB trace (inverted-F)	2.4 GHz; keep-out zone required in PCB layout

6 Locking System

6.1 Primary Lock — Solenoid Pin (Electromechanical)

As described in §3.4: spring-loaded pin, fail-locked by default, retracted by a brief solenoid pulse (GPIO + 2N7002 MOSFET, ~ 0.5 s). No continuous current draw in either state. Firmware adds a lockout after repeated failed unlock attempts (roadmap P4).

6.2 Backup — Mechanical Key Override

Recessed key slot drives a push rod retracting the pin with zero power. Required for all electronics-failure scenarios, regulatory safety, and consumer confidence.

6.3 Unlock Methods (priority order)

1. Fingerprint sensor (biometric)
2. Remote unlock via LTE-M (web dashboard)
3. BLE unlock command from paired phone
4. Physical key (mechanical override)

6.4 Auto-Lock / Dead-Man Switch

- Armed mode: the app holds a persistent GATT connection for seconds-fast disconnect detection (duty-cycled scanning remains for disarmed mode).
- Trigger: BLE supervision timeout or RSSI below a calibrated threshold (with hysteresis, as a softer trigger than hard disconnect).
- Configurable delay before alarm (default 30 s, with reconnect grace) to reduce false positives.
- Alarm fires simultaneously with lock engagement.

7 Alarm System

7.1 Audio — Piezoelectric Diaphragm + Helmholtz Chamber

Helmholtz cavity (enclosure wall):

+-----+
6 mm aperture hole -->))) sound out

Parameter	Value
Part	Same Sky CEB-35FD29
Type	Piezoelectric disc element with feedback electrode
Diameter / thickness / weight	35 mm / 0.63 mm / 3.5 g
Resonant frequency	2.9 kHz (range 2.4–3.4 kHz)
Max drive voltage	30 V peak-to-peak
Impedance	500 Ω
Termination	Wire leads: red (drive +), black (drive –/GND), blue (feedback)
Mounting	Bonded with epoxy at the rim into the enclosure acoustic cavity; not PCB-mounted; leads route through an internal channel to the PCB

+-----+	
~8 mm deep dome	<- Helmholtz cavity (~5-8 cm ³)
cavity	
+-----+	
CEB-35FD29 disc bonded	<- epoxy at disc rim (35 mm dia.)
Wire leads -> channel	<- routed to PCB
+-----+	

- Cavity depth ~8 mm; aperture ~6 mm diameter; tuned nominally to 2.9 kHz; adds ~10 mm to local enclosure depth.
- A bare piezo disc in free air produces modest SPL; the tuned cavity substantially raises effective output. SPL to be validated on the physical prototype.

7.2 Siren Drive — Boost + Full H-Bridge

Battery (3.7V) -> MT3608 boost -> 15V rail
[Full H-Bridge]
High-side: 2x BSS84 (P-ch, SOT-23, 30V)
Low-side: 2x 2N7002 (N-ch, SOT-23)
CEB-35FD29 diaphragm
+15V --> RED wire
-15V --> BLACK wire (inverted half-cycle)
= 30V p-p across element

- Drive: nRF52840 generates a 2.9 kHz square wave on two GPIOs in antiphase (complementary PWM pair).
- Feedback (blue wire): to nRF52840 ADC via voltage divider; firmware sweeps frequency to lock onto the resonance peak for maximum SPL.
- Gate resistors: 100 Ω per MOSFET gate. BSS84 body diodes provide freewheeling protection.
- Rationale: single-ended drive from 15 V yields 15 V p-p; a full bridge yields ± 15 V = 30 V p-p — exactly the rated maximum, roughly doubling acoustic output from the same supply.

7.3 Visual — Strobe LED Array

Parameter	Value
Emitter	Cree XP-E White (XPEWHT-L1-R250-00FE1), 1 W SMT, 6500 K, 70-CRI
Configuration	Array of emitters (count and placement TBD)
Drive	Battery rail via 2N7002 MOSFET; 8–10 Hz PWM strobe
Current limiting	2.2 Ω series resistor per emitter

Emitter count, parallel vs. per-channel MOSFET drive, peak battery current during strobe, and thermal handling on the enclosure are open items that must close before the PCB order.

7.4 Alarm Trigger Conditions

- BLE dead-man switch (phone out of range / disconnect, configurable delay)
- Accelerometer tamper pattern (rapid jerk / cutting vibration) via BMA400 hardware wake interrupt
- Remote trigger via LTE-M from the web dashboard
- Manual trigger via physical button (placement TBD)

Alarm response (state machine, roadmap P4): siren + strobe + lock engagement + location burst.

8 Biometric Access

Parameter	Value
Part	DFRobot SEN0348 (Mouser 426-SEN0348)
Type	Capacitive
Interface	UART → nRF52840
Template storage	80 fingerprints, on-module (no MCU RAM required)
Resolution	508 dpi
Supply	3.3 V; 60 mA active
Dimensions	21 × 5 mm
Operating temperature	−40 °C to +60 °C
FAR	<0.001 %

Capacitive sensing reads reliably on wet, dirty, or dry fingers — the principal failure mode of optical sensors in travel use. The 5 mm profile suits flush mounting on the enclosure outer face. 80 templates is ample for personal use (3–5 fingers across 1–2 users). The mechanical key remains the fallback for all biometric failure scenarios.

9 Power Architecture

Rail	Source	Consumers
3.7–4.2 V (battery)	LiPo → nPM1300 pass-through	nRF9151 VDD; MT3608 input; strobe array; solenoid (pending test)
1.8 V regulated	nPM1300 BUCK1	nRF52840 VDD; BMA400; I ² C pull-ups
15 V boost	MT3608 (C84446)	H-bridge siren supply only (30 V p-p across diaphragm)
USB-C input	nPM1300 charger	LiPo charging up to 1.5 A (PD); powers system when plugged in

Battery: LiPo, 3.7 V nominal, 800–1000 mAh, flat pouch form with integrated PCM; JST-PH 2.0 mm connector. Target: multi-day life at low LTE-M ping frequency using PSM/eDRX. The nPM1300 charger is optimised for cells up to 1000 mAh; larger cells charge at reduced C-rate. Fuel-gauge readout feeds the BLE Battery Service and location posts (currently fixed at 100 % on the Nano 33 prototype until the nPM1300 exists).

10 PCB & Passives

I²C bus: 1.8 V rail; 4.7 kΩ pull-ups to BUCK1 output; shared by BMA400 and nPM1300.

Component	Spec
32 MHz crystal	nRF52840 radio clock; ± 10 ppm; load caps per nRF52840 datasheet
32.768 kHz crystal ($\times 2$)	One per MCU; RTC and PSM timing; ± 20 ppm
BUCK inductors ($\times 2$)	$1.5\ \mu\text{H}$, $\geq 300\ \text{mA}$, low DCR, 0402/0603, for nPM1300 BUCK1/BUCK2
NTC thermistor	$10\ \text{k}\Omega$, $B=3380\ \text{K}$; battery temperature sensing for the nPM1300 charger
USB ESD protection	PRTR5V0U2X, SOT-363
MOSFETs, N-channel	2N7002, SOT-23 — H-bridge low-side, solenoid drive, strobe array drive (quantity per final array design)
MOSFETs, P-channel	$2\times$ BSS84, SOT-23, 30 V — H-bridge high-side only

11 Key Design Decisions (confirmed)

- **Zipper abandoned** — replaced by a rigid solenoid pin latch.
- **Servo cam abandoned** — solenoid pin is simpler, lighter, fail-locked, no PWM needed.
- **UHMWPE (Dyneema) abandoned** — replaced by Aramid PARATEX 400P (para-aramid, easier to source).
- **Siren drive:** full H-bridge + MT3608 boost gives 30 V p-p (vs. 15 V p-p single-ended) — doubles acoustic output.
- **Fingerprint:** SEN0348 capacitive over optical — reliable on wet/dirty fingers in travel use.
- **Backend:** FMD Server fork — working map dashboard + command dispatch infrastructure.
- **Relay architecture:** AirTag-style (phone GPS attributed to pouch) for the BLE prototype; standalone GNSS arrives with the nRF9151.
- **nRF54 rejected** — BLE Channel Sounding is useless without phone-side support; RSSI zones are the ranging ceiling. UWB would be the precision-finding route if ever needed.
- **No embedded push service** — proximity alerts are local notifications; pouch commands piggyback on the 30 s scan cycle.

12 Software & Backend

12.1 Repository Structure

```

smart_pouch/
+-- README.md          <- full technical product spec
+-- firmware/          <- device firmware (own git repo)
| +-- nrf52840/        <- PlatformIO/Arduino, Nano 33 BLE prototype (working)
| +-- nrf9151/         <- Zephyr/NCS skeleton (LTE-M/GNSS, not started)
| +-- shared/          <- ble_protocol.h - BLE UUIDs shared with Android app
+-- fmd-android/        <- fork of FMD Android (Kotlin); companion app + BLE->server relay
+-- fmd-server/         <- fork of FMD Server (Go + React); backend + web dashboard
+-- infra/             <- Ansible playbooks for Oracle Cloud deployment

```

`fmd-android` and `fmd-server` are forks of gitlab.com/fmd-foss/ projects. Upstream tracks lost Android phones; this fork adapts the stack for an IoT embedded device. All subprojects are AGPL-3.0 / GPLv3-or-later; the fork must remain open source.

12.2 Tracker Architecture (implemented — AirTag-style relay)

The Nano 33 BLE prototype has no GPS; location comes from the phone:

1. Firmware advertises BLE service `af19b3e4-d279-4a2a-9d3f-2f5e8a6bc051` under the name `SecurePouch` (same for all units). The unique unit ID (e.g. `securepouch-001`) is read from GATT characteristic `...bc052`; a status byte (LOCKED/ARMED flags, reserved) is

- ...bc053. Source of truth: `firmware/shared/ble_protocol.h`.
- 2. `BleTrackerScanService` (Android foreground service) scans 5 s on / 25 s off. On sighting, it takes the *phone's own GPS fix*, encrypts it with the pouch account's RSA public key, and POSTs it to the FMD server under the pouch's credentials (`BleTrackerRepository`; `EncryptedSharedPreferences` file `sp_ble_tracker`).
- 3. **Each pouch is its own FMD server account** — username = the BLE device UID. The web dashboard adds it as a tracker tab (device ID + that account's password).

Pairing flow (Meshtastic-style): SecurePouch Devices → Scan for Devices (`BleScanActivity`) → tap found device → GATT read of device ID → dialog asks password + registration token → register. If the username already exists on the server (HTTP 400), the app automatically falls back to logging in with the given password. Registration/login takes 30–60 s of on-device crypto (RSA keygen + Argon2) — silence is normal.

- The server's registration token gates account creation; it lives in the Ansible vault (`vault_lintu_registration_token` in `infra/host_vars/lintu-server.myvnc.com/vault.yml`).
- `securepouch-001` is registered as the dev test unit.
- Access tokens expire after **7 days**; refresh logic is not yet implemented on either client (top P1 item).

12.3 fmd-server (Go + React)

Language	Go 1.24
Database	SQLite via GORM (embedded; no separate DB process)
Frontend	React 19, TypeScript, Vite, Tailwind CSS v4, shadcn/ui, Leaflet maps, Zustand
Package manager	pnpm (≥ 10)
Key libraries	zerolog, spf13/cobra+viper, Prometheus metrics
Distribution	Single Go binary embedding the compiled React frontend

API structure (backend/): `api.go` (HTTP mux, security-headers middleware), `apiv1.go` (REST v1 handlers: location, command, picture, salt / `requestAccess` / `pubKey` endpoints), `backend.go` (server setup).

Dashboard fork changes (implemented):

- Multi-device map: phone + tracker tabs (`SidePanel`), per-device colours, eye visibility toggles.
- Time filter (1h/6h/24h/7d/All) + fit-all control; history scrubber with pan/highlight (`LocationMap`).
- Tracker credentials: session token in `localStorage['fmd-trackers']`; RSA keys in IndexedDB (`tracker:{deviceId}`). Tracker fetches use raw `fetch` so a tracker 401 never logs out the main account.
- Dark-mode fixes in `ui/native-select.tsx` (`color-scheme: dark` + explicit option colours).

Fork changes still pending:

- Command types `lock`, `unlock`, `arm`, `disarm`, `alarm` (do not exist upstream).
- Dashboard lock/alarm controls; remove or stub Android-only features (camera capture, factory reset).

12.4 fmd-android (Kotlin)

- Standard Android Gradle project; `dev` flavour installs in parallel with `prod`; keyword prefix `fmdev`.

- SecurePouch additions live in `de.nulide.findmydevice.securepouch` (repository, discovered-device model) and `ui/settings/` (`PouchManagementActivity`, `BleScanActivity`, `BleScanViewModel`).
- Push: `UnifiedPush` (user runs `Sunup` as distributor) for FMD phone commands only; no embedded push service.

12.5 firmware/nrf52840 (PlatformIO, Arduino framework)

- Target `nano33ble`; library `ArduinoBLE`.
- Advertises the SP service; OS-pairable (Just Works); GAP name + Generic Tag appearance; standard Battery Service (fixed 100% until the `nPM1300` exists); connect LED.
- Per-unit `DEVICE_UID` constant must be changed before flashing each unit.
- `nRF52840` is BLE-only (no Classic); Android system-settings pairing works but the app's GATT flow does not require it.

12.6 infra (Ansible) / Deployment

```
nRF9151 / Android relay --HTTPS--> FMD Server (Go binary, port 8080)
                                   |
                                   Caddy reverse proxy (auto-TLS, Let's Encrypt)
                                   |
                                   Browser dashboard (HTTPS required by WebCrypto)
```

Provider / tier	Oracle Cloud Infrastructure, Always Free (ARM Ampere A1, Ubuntu 22.04)
Live host	<code>lintu-server.myvnc.com</code> ; hosts named NATO-phonetically from Foxtrot
Stack	Docker Compose; Caddy (auto-TLS); FMD Server container
Build policy	Builds happen on the laptop, never the server (cross-compile + upload zip)
Secrets	Vault password at <code>~/.ansible/vault-pass</code> ; SSH key outside repo; <code>vault.yml</code> files always ansible-vault encrypted — never commit secrets unencrypted
Backup hosting	Fly.io (Docker-native PaaS, auto-HTTPS on <code>.fly.dev</code>)

12.7 Build & Deploy Commands

```
# Web dashboard + server (deploy pipeline)
cd fmd-server/web && pnpm install && pnpm run build && cd ..
GOOS=linux GOARCH=amd64 go build -o fmd-server-amd64 .
GOOS=linux GOARCH=amd64 go build -o fmd-server-ctl-amd64 ./ctl/
zip -j dist/fmd-server-securepouch-v0.1.0.zip fmd-server-amd64 fmd-server-ctl-amd64
cd ../infra
ansible-playbook 40-securepouch-server.yml \
  --vault-password-file ~/.ansible/vault-pass
# then hard-refresh the dashboard (Ctrl+Shift+R)

# Local dev server (frontend must be built first)
cd fmd-server && go run . serve

# fmd-android
cd fmd-android && ./gradlew assembleDevDebug
adb install -r app/build/outputs/apk/dev/debug/app-dev-debug.apk
# relay debugging:
adb logcat -s BleTrackerRepository BleTrackerScanService Volley

# firmware (Nano 33 BLE; pio is not on PATH)
cd firmware/nrf52840
~/platformio/penv/bin/pio run -t upload
~/platformio/penv/bin/pio device monitor    # 115200 baud
```

13 Manufacturing

Prototype run: 5 fully assembled PCBs from JLCPCB (fabrication + SMT assembly in one order). Consigned parts (nRF9151, nRF52840, nPM1300, Cree LEDs) shipped to the JLCPCB warehouse ahead of the assembly order.

Supplier	Role
JLCPCB	PCB fabrication + full SMT assembly; 5-unit prototype run
Circuitos Impresos 2CI (Barcelona)	Quick-turn bare PCBs for rapid iteration (from 4 h turnaround)
Hologram	eUICC SIM provisioning and global data plan
Mouser	Consigned parts and off-PCB modules

Assembly notes:

- nRF9151 (LGA), nRF52840 (QFN73), nPM1300 (WLCSP/QFN): precision reflow required; not hand-solderable.
- BLE trace antenna requires a copper keep-out zone — verify in DFM before ordering.
- FXP301 antenna attaches via adhesive + 2× U.FL connectors post-assembly.
- SEN0348 connects via header; flush-mounted on the enclosure outer face.
- CEB-35FD29 mounts in the enclosure Helmholtz cavity; wire leads route through an internal channel to the PCB.
- Gen-1 demonstrator enclosure is FDM-printed; final enclosure expected to be CNC-machined aluminium, form factor TBD.

14 Roadmap / TODO

Ordered by priority; each phase has a definition of done (DoD), and the next phase does not start until the previous DoD is met. Status as of 2026-06-11.

14.1 P0 — Working tracker demo (complete; walk test pending)

DoD: walk around the block with the pouch in a bag and see the track on the dashboard.

- ☒ Flash updated firmware to Nano 33 BLE
- ☒ Install rebuilt Android app
- ☒ Deploy server (map features + dark-mode fix)
- ☒ Pair pouch in app (auto-login fallback verified 2026-06-11)
- ☒ Verify relay: logcat shows `Location posted for pouch 'securepouch-001'`
- ☒ Dashboard: tracker tab added
- ☐ End-to-end walk test; check history scrubber + time filter against reality

14.2 P1 — Demo hardening

- ☐ **Token refresh (Android):** pouch access tokens expire after 7 days. `BleTrackerRepository` stores `hashedPw`, so it can re-run `requestAccess` on a 401 and retry the post. Without this the relay dies silently after a week.
- ☐ **Token refresh (web):** tracker sessions also expire after 7 days; the dashboard should prompt to re-enter the tracker password instead of showing nothing.
- ☐ Phone-side “pouch left behind” notification — app already tracks `lastSeen`; raise a local notification when a pouch is unseen for N minutes.
- ☐ Real RSSI logging — attach RSSI to each posted sighting (groundwork for the distance ring).
- ☐ Multi-unit sanity check: flash `securepouch-002` on a second board; confirm separate tabs/colours.

14.3 P2 — Anti-theft feature set

- ☐ Armed/disarmed state end to end: app toggle → GATT write to STATUS characteristic (SP_STATUS_ARMED flag already reserved in `ble_protocol.h`)
- ☐ Persistent GATT connection while armed (app); duty-cycled scan stays for disarmed mode
- ☐ Pouch-side dead-man alarm (firmware): on supervision timeout while armed, wait configurable delay (default 30s with reconnect grace) → alarm. Built-in LED + serial as siren stand-in on the Nano 33.
- ☐ RSSI radius fence: calibrated threshold + hysteresis as a softer trigger than hard disconnect
- ☐ Distance ring on dashboard: pouch rendered as a translucent circle around the phone marker, radius from RSSI
- ☐ Server command types: `lock`, `unlock`, `arm`, `disarm`, `alarm`
- ☐ Dashboard arm/disarm/alarm controls per tracker tab
- ☐ App relays server commands to pouch over BLE, piggybacking on the 30s scan cycle

14.4 P3 — nRF9151 DK prototype (standalone, no phone)

Order the DK and activate the Hologram SIM early — both have lead time.

- ☐ nRF Connect SDK / Zephyr toolchain (skeleton in `firmware/nrf9151/`)
- ☐ Hologram SIM: pick data plan tier, activate, verify LTE-M attach on DK
- ☐ GNSS fix acquisition (+ A-GNSS via nRF Cloud)
- ☐ FMD API client in C: registration/login (Argon2 → `/salt`, `/requestAccess`); location encryption (RSA-OAEP + AES-256-GCM hybrid, mbedTLS) matching what the dashboard decrypts; HTTPS POST to `/location`; token refresh on 401
- ☐ Command polling loop (`/command`) → `lock/unlock/arm/disarm/alarm`
- ☐ UART protocol nRF9151 ↔ nRF52840: message format (status, commands, GNSS hand-off); document in `firmware/shared/`
- ☐ Port nRF52840 firmware from Arduino to Zephyr *before* PCB bring-up, while the Nano 33 still works as fallback
- ☐ Power profile on DK: sleep currents, GNSS duty cycle, LTE-M PSM/eDRX tuning

14.5 P4 — Peripheral firmware (write before wiring)

Goal: when the PCBs arrive, bring-up is “flash and probe,” not “write drivers while holding a multimeter.” All developable on the Nano 33 / DK with breakout modules.

- ☐ BMA400 driver (I²C) + motion-wakeup interrupt + tamper pattern detection
- ☐ SEN0348 driver (UART): enroll / verify / delete flows
- ☐ Solenoid lock driver: single-pulse GPIO via MOSFET, fail-locked logic, lockout after repeated failed attempts
- ☐ Siren driver: 2.9kHz complementary PWM pair for the H-bridge + resonance lock via feedback electrode (ADC)
- ☐ Strobe driver: 8–10Hz PWM on the Cree XP-E array
- ☐ nPM1300 config: charge profile, fuel gauge → real battery level in BLE Battery Service + location posts
- ☐ Alarm state machine: triggers (dead-man, tamper, remote, button) → siren + strobe + lock + location burst

14.6 P5 — Hardware / mechanical (parallel track)

- ☐ Solenoid part selection + voltage testing (3.7V rail vs. 5V boost)
- ☐ Final enclosure form factor decision; CNC machining (5 prototype units)
- ☐ Strobe array definition: emitter count, placement, drive topology, peak current and thermal budget
- ☐ Acoustic cavity SPL validation on physical prototype

- ☐ Physical button placement (alarm trigger / manual override)
- ☐ PCB DFM check: BLE trace antenna keep-out before JLCPCB order
- ☐ JLCPCB order: 5 assembled PCBs; consigned parts shipped to warehouse ahead of assembly
- ☐ Extended size variant: dimensions and gusset design
- ☐ Hologram data plan tier selection (based on ping frequency)